

# Mingyun Kang

Researcher in Construction Management & Automation

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## Education

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### Seoul National University of Science and Technology

*B.S. in Architectural Engineering, School of Architecture*

*Seoul, Republic of Korea*

*Mar 2017 – Feb 2024*

- Double Major: Business Administration
- GPA : 3.8 / 4.5

### Seoul National University of Science and Technology

*M.S. in Architectural Engineering (Integrated Bachelor-Master Program)*

*Seoul, Republic of Korea*

*Mar 2024 – Aug 2025*

- GPA : 4.22 / 4.5
- Thesis : Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System (Advisor : Prof. Taehoon Kim)

## Research Experience

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### Research Assistant

*CONICT Lab, Seoul National University of Science and Technology*

*Seoul, Republic of Korea*

*Dec 2021 – Aug 2025*

Conducted research on computer vision-based automated construction supervision, 3D reconstruction, and human-robot collaboration at construction sites.

- Published 3 SCIE journal papers and 2 SCOPUS conference papers as first/co-author
- Filed 5 domestic patent applications related to construction automation and vision-based inspection
- Participated in 3 national R&D projects funded by NRF and the Ministry of Land, Infrastructure and Transport (MOLIT)

### Post-Master's Researcher

*Korea Institute of Civil Engineering and Building Technology (KICT)*

*Goyang, Republic of Korea*

*Nov 2025 – present*

Post-Construction Evaluation Center — Systematically evaluating outcomes of publicly funded SOC

- Compliance & Implementation Support: Developing research frameworks to encourage clients to fulfill post-construction evaluation requirements and to facilitate smooth reporting workflows.
- Feedback Loop & Knowledge Management: Designing mechanisms to extract structured insights from collected evaluation data, enabling lessons learned from completed projects to feed back into future initiatives and build a virtuous cycle of construction knowledge.

## R&D Projects

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### Digital Mapping-based Platform Research Laboratory for Healthy Construction Lifecycle

*June 2022 – Feb 2025*

National R&D project exploring digital mapping-based platforms for the full construction lifecycle. Focused on computer vision applications for automated construction supervision.

### Development of Digital-based Architectural Supervision and Construction Automation Robot Technology

*Apr 2022 – Dec 2026*

Large-scale national R&D project (KRW 24,490M) funded by MOLIT aimed at developing digital architectural supervision systems and construction automation robots.

### Development of Human-Robot Collaboration Technology and Integrated Operation System for High-Altitude Work at Construction Sites

*Apr 2025 – Dec 2027*

Ongoing national R&D project (KRW 17,000M) funded by MOLIT developing XR-based human-robot collaboration platforms for multipurpose high-altitude construction work.

## Publications (SCIE)

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- Computer Vision-Based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System** Dec 2024  
Applied Sciences, 15(1), 125.  
**Mingyun Kang**, Sebeen Yoon, Taehoon Kim  
[10.3390/app15010125](https://doi.org/10.3390/app15010125)  (Applied Sciences)
- Approach to Enhancing Panoramic Segmentation in Indoor Construction Sites Based on a Perspective Image Segmentation Foundation Model** Apr 2025  
Applied Sciences, 15(9), 4875.  
Juho Han, Sebeen Yoon, **Mingyun Kang**, Taehoon Kim  
[10.3390/app15094875](https://doi.org/10.3390/app15094875)  (Applied Sciences)
- Automated Vision-based Location Monitoring for Jack Support** Jan 2025  
KSCE Journal of Civil Engineering, 100374.  
Sebeen Yoon, **Mingyun Kang**, Minhyung Kim, Taehoon Kim  
[10.1016/j.kscej.2025.100374](https://doi.org/10.1016/j.kscej.2025.100374)  (KSCE Journal of Civil Engineering)

## Publications (SCOPUS)

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- Instance Segmentation of Exterior Insulation Finishing System using Synthetic Datasets** June 2024  
ISARC Vol. 41, pp. 1176-1181. IAARC Publications.  
**Mingyun Kang**, Sebeen Yoon, Juho Han, Sanghyeon Na, Taehoon Kim  
[10.22260/ISARC2024/0152](https://doi.org/10.22260/ISARC2024/0152)  (ISARC 2024 — International Symposium on Automation and Robotics in Construction)
- Comparative Study of Structure from Motion on Construction Site** July 2025  
ISARC Vol. 42, pp. 1395-1400. IAARC Publications.  
**Mingyun Kang**, Sangmin Lee, Sebeen Yoon, Taehoon Kim  
[10.22260/ISARC2025/0180](https://doi.org/10.22260/ISARC2025/0180)  (ISARC 2025 — International Symposium on Automation and Robotics in Construction)

## Publications (International Conference)

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- Computer Vision-Based Tile Counting Model for Automated Progress Monitoring** June 2023  
CCC 2023, Keszthely, Hungary.  
**Mingyun Kang**, Wooseok Lee, Junsang Yoo, Sebeen Yoon, Taehoon Kim  
(Creative Construction Conference (CCC 2023))
- A Preliminary Study on Automatic Jack Support Installation Interval Measurement Model** June 2023  
CCC 2023, Keszthely, Hungary.  
Sebeen Yoon, **Mingyun Kang**, Taehoon Kim  
(Creative Construction Conference (CCC 2023))
- Approach for Improving Panoramic Image Segmentation Quality on Construction Sites using SAM** June 2024  
CCC 2024, Prague, Czech Republic.  
Juho Han, Sanghyeon Na, **Mingyun Kang**, Sebeen Yoon, Taehoon Kim  
(Creative Construction Conference (CCC 2024))

## Publications (Domestic)

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- Computer Vision-based Adhesion Quality Inspection Automation Model for Exterior Insulation and Finishing System** Mar 2023  
JKIBC, 23(2), 165-173. (KCI)  
Sebeen Yoon, **Mingyun Kang**, Hyeonseung Jang, Taehoon Kim  
(Journal of the Korean Institute of Building Construction (JKIBC))
- Applicability Enhancement of Deep Learning-based Computer Vision Models for 360° Panoramic Images** Nov 2023  
KICEM 2023, Goseong, 205-206.  
**Mingyun Kang**, Junsang Yoo, Wooseok Lee, Taehoon Kim  
(Proceedings of KICEM Annual Conference)
- Performance Comparison Study on Image-based 3D Space Generation** Oct 2024  
AIK Fall 2024, Gyeongju, 974-975.  
**Mingyun Kang**, Sebeen Yoon, Taehoon Kim  
(Architectural Institute of Korea Fall Conference 2024)
- A Basic Study on 3D Reconstruction Performance Improvement Using 360° Camera** Nov 2024  
KIBC Autumn 2024, 203-204.  
**Mingyun Kang**, Juho Han, Sebeen Yoon, Taehoon Kim  
(Proceedings of KIBC Autumn Conference)

## Patents

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- Automatic Supervision Method and Device for Adhesive Application Quality** Apr 2023  
Domestic Patent Application No. 10-2023-0053756. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.  
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**

## Patents (Pending)

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- Automated Construction Site Supervision System** Sept 2023  
Domestic Patent Application No. 10-2023-0125464. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.  
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**
- Tile Quantity Counting System** Oct 2023  
Domestic Patent Application No. 10-2023-0135013. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.  
Taehoon Kim, Sebeen Yoon, **Mingyun Kang**, Wooseok Lee, Junsang Yoo
- Method for Measuring Rebar Spacing Based on Single Image Information** Sept 2024  
Domestic Patent Application No. 10-2024-0127210. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.  
Taehoon Kim, **Mingyun Kang**, Cheolhun Park, Jihyeon Lee, Jieun Lee, Yunjae Jeon
- Method for Detecting Illegal Rooftop Buildings Based on Video Image Analysis** Aug 2025  
Domestic Patent Application No. 10-2025-0117608. Seoul National University of Science and Technology Industry-Academic Cooperation Foundation.  
Taehoon Kim, **Mingyun Kang**, Seonwoo Lee, Yeon Choi, Yujin Oh, Gangmin Bae

## Awards

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### Best Paper Award

May 2023

Awarded at the Spring Conference 2023 for the paper: A Basic Study on Image-based Numerical Measurement Model for Construction Sites  
Korean Institute of Building Construction (KIBC)

### Best Paper Award

May 2024

Awarded at the Spring Conference 2024 for the paper: A Basic Study on Rebar Spacing Measurement Method Based on Single Image Information  
Korean Institute of Building Construction (KIBC)

### Excellence Paper Award

May 2025

Awarded at the Spring Conference 2025 for the paper: A Basic Study on Image Analysis-based Crackdown on Illegal Buildings on Rooftops  
Korean Institute of Building Construction (KIBC)

### Outstanding Thesis Award

Aug 2025

Awarded for the Master's thesis: Computer Vision-based Adhesion Quality Inspection Model for Exterior Insulation and Finishing System  
Seoul National University of Science and Technology

## Languages & Skills

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◦ Korean (Native) · English (Advanced) · Python (Advanced)

## Certificates

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Architectural Engineer License

*June 2025*

TOEFL iBT — Score 4.5/6.0 (90/120)

*July 2026*

RC 5.0 / LC 5.0 / WRT 3.5 / SPK 4.5

## References

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Professor Taehoon Kim